

SIVAKUMAR RAMAKRISHNAN

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SUMMARY

Applied AI engineer who ships production agentic systems and automation end to end, from AWS backends and SQL data models to Claude-powered multi-agent workflows that turn days of manual work into minutes. Daily user of AI-assisted development (Claude Code); comfortable owning problems from idea to deployment.

PROFESSIONAL EXPERIENCE

RICOH - Senior Applied AI Engineer (Full-time) May 2026 - Present

- Architected and shipped a Claude-powered multi-agent platform: an orchestrator that routes requests to a Data Analyst agent and a Service agent, which retrieves maintenance documentation to help field technicians diagnose and fix printer issues.
- Added image support to the Service agent, pairing step-by-step repair guidance with procedural illustrations.
- Built a high-throughput document pipeline on Ray, processing thousands of documents in under an hour.
- Automated localized UI snapshot creation with an agentic harness (130 screenshots in each of 7 languages, 900+ images), cutting turnaround from 60 hours of manual work to just 40 minutes.

RICOH - Applied AI Engineer (Internship) May 2025 - Apr 2026

- Led the Service agent's initial deployment, adding human feedback loops for continuous improvement, conversational memory, and tool calling to JIRA and Redmine via AWS connectors.
- Designed the foundational AWS architecture (Lambda, DynamoDB, S3, Aurora SQL) as a shared platform for future agents.

ZOHO - Data Scientist, Research and Development May 2022 - Jul 2024

- Architected a scalable RAG system serving 10K+ daily queries, cutting response latency by 20% and hallucinations by 35% through hybrid retriever-reranker fusion and vLLM.
- Built customer-assistance AI with PyTorch and Transformers, reaching 90% accuracy in phishing detection and parsing 2K+ resumes per month.

SIEMENS - Intern Apr 2021

- Studied PET-CT and MRI imaging workflows and prototyped a CNN classifier on public CT scan data to explore how AI could help flag abnormal scans for faster radiologist review.

SELECTED PROJECTS

- **Edge AIoT for Product Inspection:** Built a full-stack inspection system (CNN, Flask, AWS DynamoDB, Raspberry Pi) that identifies defective microchips in real time with 92% accuracy. [Demo](#)
- **Invoice Image to JSON (PaliGemma):** Developed a document-extraction pipeline that converts invoices across 200+ layouts into structured JSON with 88% accuracy. [Model](#)
- **Goodreads Rating Prediction:** Won 1st place in a Kaggle competition with a fine-tuned T5 model and custom preprocessing (0.70 mean F1). [Code](#)

KEY SKILLS

AI & Agents: Claude Code, multi-agent orchestration, tool calling, RAG, LLM evaluation, NLP, computer vision, multimodal AI

Programming & Data: Python, SQL, PostgreSQL, JavaScript, React, Java, R, relational data modeling, REST APIs, FastAPI, Streamlit

Cloud & Tools: AWS (Lambda, DynamoDB, S3, Aurora), Docker, CI/CD, Git, PyTorch, Pandas, Ray, PySpark, Power BI, GCP, Azure

EDUCATION

Master of Science in Data Science Aug 2024 - May 2026
University of Colorado Boulder GPA: 4.00/4.00

Bachelor of Engineering in Electronics and Communication Aug 2018 - Jun 2022
Sri Sai Ram Engineering College, Chennai, India GPA: 3.65/4.00

Certifications: [NLP Specialization](#), [Deep Learning Specialization](#), [Machine Learning](#) (Coursera)